Enrollment No._____

GANPAT UNIVERSITY

B.Sc. (CA & IT) SEM-III (CBCS)

APRIL-JUNE 2024 Remedial

U13A1DS: DATA STRUCTURE

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy Pattern**
1	Write down algorithm for circular queue operations.	06	CO1	BTL2
2	Write down algorithm for merge link lists.	06	CO3	BTL2
3	Explain Threaded tree.	06	CO2	BTL5
4	Explain the importance of stack machine with example.	06	CO1	BTL6
5	Write down the algorithms for all stack operations.	06	CO3	BTL3
6	Explain binary search with example.	06	CO4	BTL6
7	Explain Insertion sort with suitable example.	06	CO4	BTL6
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy Pattern**
1	What is data structure? Explain the types of it.	06	CO2	BTL1
2	Write down algorithm for Deque operations.	06	CO3	BTL3
3	Write down algorithm to insert a node at the beginning of the doubly link list.	06	CO3	BTL2
Q.3	**Answer the following: (Any Two)**	**12**		
1	Write down algorithm to insert the node in tree	06	CO3	BTL3
2	Explain different tree traversal method with example	06	CO4	BTL6
3	Explain sequential search with example and algorithm	06	CO4	BTL5
Q.4	**Answer the following: (Any One)**	**06**		
1	Explain selection sort with suitable example.	06	CO4	BTL4
2	Explain bubble sort with example.	06	CO4	BTL6

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc.(CA&IT) Sem-III (CBCS)

APRIL-JUNE 2024 Remedial

U13A2OCP: OBJECT ORIENTED CONCEPTS AND PROGRAMMING

Time: 3 Hours Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy Pattern**
1	Explain JDK, JRE, JVM.	06	CO1	BTL3
2	What is Polymorphism? Explain its types in detail.	06	CO2	BTL4
3	Explain thread life cycle and Thread Priorities with example.	06	CO2	BTL5
4	What is generic? Explain bounded types, and generic hierarchies in detail.	06	CO3	BTL4
5	How to implements lambda expression in java?	06	CO4	BTL6
6	Which are the access modifiers in java? Explain for class and package.	06	CO3	BTL5
7	Explain in detail Fork/Join using parallel programming.	06	CO5	BTL4
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy Pattern**
1	List out any 10 methods of String handling class with return type, parameters of the method and use of it.	06	CO1	BTL3
2	What is Inheritance? Explain its supported types with example.	06	CO2	BTL4
3	Illustrate the use of static keyword with suitable example.	06	CO1	BTL5

Q.3	**Answer the following: (Any Two)**	**12**		
1	Explain block expression and generic function interface in detail.	06	CO4	BTL5
2	Explain Map Interface in detail.	06	CO3	BTL6
3	Discuss the use of method and constructor references with suitable case.	06	CO4	BTL4
Q.4	**Answer the following: (Any One)**	**06**		
1	Explain in detail CountDownLatch, CyclicBarrier, Semaphore, Exchanger, Phaser.	06	CO5	BTL3
2	Discuss Synchronization in Java?	06	CO5	BTL3

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B. Sc. (CA & IT) – III (CBCS)

APRIL-JUNE 2024 Remedial

U13A3DMS: Database Management System

Time: 3 Hours Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
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4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy Pattern**
1	What is DBMS? Discuss benefits of DBMS.	06	CO1	BTL3
2	Explain External level and Internal level of Database System Architecture.	06	CO1	BTL2
3	Write a short note on 5NF.	06	CO2	BTL5
4	What is SQL? Describe and analyze the components of SQL.	06	CO3	BTL4
5	Differentiate Delete Vs. Truncate command.	06	CO3	BTL5
6	What is scalar function? Illustrate any six string function with proper example.	06	CO4	BTL6
7	Discuss brief about Index object in oracle.	06	CO5	BTL4
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy Pattern**
1	What is Data Model? Demonstrate Hierarchical Data Model in detail.	06	CO1	BTL6
2	Differentiate DBMS Vs. RDBMS.	06	CO1	BTL5
3	Illustrate E/R Diagram in details.	06	CO2	BTL4

Enrollment No._____

GANPAT UNIVERSITY

B. Sc. (CA & IT) – III (CBCS)

APRIL-JUNE 2024 Remedial

U13A3DMS: Database Management System

Time: 3 Hours Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
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5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy Pattern**
1	What is DBMS? Discuss benefits of DBMS.	06	CO1	BTL3
2	Explain External level and Internal level of Database System Architecture.	06	CO1	BTL2
3	Write a short note on 5NF.	06	CO2	BTL5
4	What is SQL? Describe and analyze the components of SQL.	06	CO3	BTL4
5	Differentiate Delete Vs. Truncate command.	06	CO3	BTL5
6	What is scalar function? Illustrate any six string function with proper example.	06	CO4	BTL6
7	Discuss brief about Index object in oracle.	06	CO5	BTL4
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy Pattern**
1	What is Data Model? Demonstrate Hierarchical Data Model in detail.	06	CO1	BTL6
2	Differentiate DBMS Vs. RDBMS.	06	CO1	BTL5
3	Illustrate E/R Diagram in details.	06	CO2	BTL4

Q.3	**Answer the following: (Any Two)**	**12**		
1	List out types of Constraints. Explain check constraint with proper example.	06	CO3	BTL3
2	Demonstrate oracle group functions with example.	06	CO4	BTL5
3	Discuss brief about oracle sub query.	06	CO4	BTL4
Q.4	**Answer the following: (Any One)**	**06**		
1	Explain various options of SET commands.	06	CO5	BTL2
2	Discuss Grant and Revoke command with example.	06	CO5	BTL3

**** End of the Paper ****

GANPAT UNIVERSITY

Format for Review of Question Paper

Date of Examination:

Examination Cycle:

Semester:

Faculty:

Program:

Subject with Code No:

Review Parameter	Marks out of 10 for each parameter	Remark, if any	Marking by Exam Audit Committee
Format of Question Paper			
Clarity of language and Spelling mistakes			
Coverage of syllabus as per weightage			
Repetition of questions/Incomplete questions/Out of syllabus questions			
Inclusion of Innovative/Hard /Average/Easy questions as per need			
Mapping with Course outcomes & Bloom Taxonomy level			

Total Marks out of 60				
Excellent	Very Good	Good	Average	Poor
More than 57	53-57	48-52	42-47	Less than 42

Name and signature of Reviewer: (1)

(2)

- **Any immediate attention required to bring into the notice of University officer? If Yes, please mention:**

	Signature of reporting Faculty/Staff

- **Decision Taken:**

	Authorized Signature

- **Guideline for Question Paper Evaluation**
  - ✓ The Question Paper is to be evaluated in first half an hour of the examination by Faculty/Teacher concern to the subject.
  - ✓ Any major query in the question paper is to be reported to Centre coordinator/Principal/Dean, which may forward to Registrar/Controller of Examination. The remedial action required if any, will be communicated to Centre coordinator.
  - ✓ The filled form is to be submitted to Centre coordinator/Principal, which will be utilized for Examination audit process.

Enrollment No._____

GANPAT UNIVERSITY

B.Sc CA&IT – Semester -III

APRIL-JUNE 2024 Remedial

U13A5OR OPERATION RESEARCH

Time: 3 Hours Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
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5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I

Q-1	Answer the following. (Any 5)	Marks 30	CO	Bloom's Taxonomy Pattern
1	A company has two plants, each of which produces and supplies two products: A and B. The plants can each work up to 16 hours a day. In plant 1, it takes three hours to prepare and pack 1,000 gallons of A and one hour to prepare and pack one quintal of B. In plant 2, it takes two hours to prepare and pack 1,000 gallons of A and 1.5 hours to prepare and pack a quintal of B. In plant 1, it costs Rs 15,000 to prepare and pack 1,000 gallons of A and Rs 28,000 to prepare and pack a quintal of B, whereas in plant 2 these costs are Rs 18,000 and Rs 26,000, respectively. The company is obliged to produce daily at least 10 thousand gallons of A and 8 quintals of B. Formulate this problem as an LP model to find out as to how the company should organize its production so that the required amounts of the two products be obtained at the minimum cost.	06	CO1 & CO2	BTL1, BTL2, BTL6
2	Solve the LPP by using Simplex method for the optimal solution:	06	CO1 & CO2	BTL1, BTL2, BTL6

Maximize $Z = x_1 - 3x_2 + 2x_3$

Subject to constraints $3x_1 - x_2 + 3x_3 \leq 7$,

$-2x_1 + 4x_2 \leq 12$,

$-4x_1 + 3x_2 + 8x_3 \leq 10$ and

$x_1, x_2, x_3 \geq 0$.

3 Determine an initial basic feasible solution to the following transportation problem by using VAM. 06 CO2 & CO3 BTL3, BTL4, BTL5

Destination/source	D1	D2	D3	D4	Supply
S1	21	16	15	3	11
S2	17	18	24	23	13
S3	32	27	10	41	19
Demand	6	6	8	23	

4 Define balanced transportation problem. Determine an initial basic feasible solution to the following transportation problem by using Least Cost Method 06 CO2 & CO3 BTL3, BTL4, BTL5

D/S	D1	D2	D3	D4	Supply
S1	11	13	17	14	250
S2	16	18	14	10	300
S3	21	24	13	10	400
Demand	200	225	275	250	

5 Solve the transportation problem and obtained the optimal solution. (Using MODI method) 06 CO2 & CO3 BTL3, BTL4, BTL5

D/S	D1	D2	D3	D4	Supply
S1	2	2	2	1	3
S2	10	8	5	4	7
S3	7	6	6	8	5
Demand	4	3	4	4	

6 06 CO4 BTL4, BTL5, BTL6

		Job I	II	III	IV
Men	A	42	35	28	21
	B	30	25	20	15
	C	30	25	20	15
	D	24	20	16	12

Find out how men should be assigned the jobs in way that will minimize the total time taken.

7 A book binder has one printing press, one binding machine and manuscripts of 7 different books. The times required for performing printing and binding operations for different books are shown below: 06 CO5 BTL5, BTL6

Book	: 1	2	3	4	5	6	7
Printing time (hours)	: 20	90	80	20	120	15	65
Binding time (hours)	: 25	60	75	30	90	35	50

Section – II

Q-2	Answer the following (Any Two)	Marks 12	CO	Bloom's Taxonomy Pattern
1	Use the graphical method to solve the following LP problem. Maximize $Z = 2x_1 + x_2$ Subject to constraints $x_1 + 2x_2 \leq 10$, $x_1 + x_2 \leq 6$, $x_1 - x_2 \leq 2$, $x_1 - 2x_2 \leq 1$ and $x_1, x_2 \geq 0$.	06	CO1 & CO2	BTL1, BTL2, BTL6
2	Solve the LPP by using Simplex method for the optimal solution: Maximize $Z = x_1 - 3x_2 + 2x_3$ Subject to constraints $3x_1 - x_2 + 3x_3 \leq 7$, $-2x_1 + 4x_2 \leq 12$, $-4x_1 + 3x_2 + 8x_3 \leq 10$ and $x_1, x_2, x_3 \geq 0$.	06	CO1 & CO2	BTL1, BTL2, BTL6
3	Determine an initial basic feasible solution to the following transportation problem by using NWCR.	06	CO2 & CO3	BTL3, BTL4, BTL5

D/S	D1	D2	D3	D4	D5	Supply
S1	5	1	8	7	5	15
S2	3	9	6	7	8	25
S3	4	2	7	6	5	42
S4	7	11	10	4	9	13
Demand	30	20	15	10	20	

Q.3	Answer the following: (Any Two)	12		
1	Solve the transportation problem to maximize the profit.	06	CO2 & CO3	BTL3, BTL4, BTL5

D/S	D1	D2	D3	D4	Supply
S1	15	51	42	33	23
S2	80	42	26	81	44
S3	90	40	66	60	33
Demand	23	31	16	30	

2	Solve the transportation problem and obtained the optimal solution. (Using MODI method)	06	CO2 & CO3	BTL3, BTL4, BTL5

D/S	D1	D2	D3	Supply
S1	5	1	8	12
S2	9	4	0	14
S3	17	6	7	4
Dem.	9	10	11	

3	**Job**	06	CO4	BTL4,

BTL5, BTL6

		I	II	III	IV	V
Typist	A	85	75	65	125	75
	B	90	78	66	132	78
	C	75	66	57	114	69
	D	80	72	60	120	72
	E	76	64	56	112	68

Find out how typist should assign a job so that will minimize the cost.

Q.4 Answer the following: (Any One) 06

1 A manufacturing company processes 6 different jobs on two machines A and B. Number of units of each job and its processing times on A and B are given below. Find the optimal sequence, the total minimum elapsed time and idle time for either machine. 06 CO5 BTL5, BTL6

Job Number	*Number of Units of Each Job*	*Processing Time (in Minutes)*	
		Machine A	*Machine B*
1	3	5	8
2	4	16	7
3	2	6	11
4	5	3	5
5	2	9	7.5
6	3	6	14

2 We have six jobs, each of which must go through machines A, B and C in the order ABC. Processing time (in hours) are given in the following table: 06 CO5 BTL5, BTL6

Job	:	1	2	3	4	5	6
Machine A	:	8	3	7	2	5	1
Machine B	:	3	4	5	2	1	6
Machine C	:	8	7	6	9	10	9

Determine a sequence for the five jobs that will minimize the elapsed time t.

**** End of the Paper ****